

viii. The Petroleum

Petroleum is a liquid, formed as a result of different mixtures of hydrocarbons. Petroleum is a multicolored substance. The multicolours are due to the mixture of different chemicals present in it. Petroleum is found in sedimentary rocks. It gets accumulates in the stoma and interstices of porous sandstones and limestones.

*The first commercial oil well was drilled in Romania in 1857. Romania is the first country in the world whose crude oil production has an official document (1857 AD. 275 tonnes) and was recorded at the international level. However according to the Ministry of Petroleum & Natural Gas, GOI Edwin L. Drake drilled the worlds first oil well in 1859 at Titusville, Pennsylvania, U.S.A. According to Indian Minerals Year Book 2022, the largest share of reserves of world crude oil is available in Middle East (48.3%), followed by South & Central America (18.7%), North America (14%), CIS (8.4%), Africa (7.2%), Asia Pacific (2.6%) and Europe (0.8%). *According to IMBY-2022 the top five leading countries in terms of crude oil proved reserves are – **Venezuela** (48.0 thousand tonnes), **Saudi Arabia** (40.9 thousand tonnes), **Canada** (27.1 thousand tonnes), **Iran** (21.7 thousand tonnes) and **Iraq** (19.6 thousand tonnes). *According to World Mineral Production, 2018-22 (data 2022) the largest crude oil producers in the world are – **USA** (759.50 million tonnes), **Saudi Arabia** (573.10 million tonnes), **Russia** (548.50 million tonnes), **Canada** (274.00 million tonnes), **Iraq** (221.30 million tonnes).

*The important petroleum producing countries in Asia are – Saudi Arabia, Iran, Iraq, UAE, Kuwait, Oman, Qatar etc. *The largest oil-producing country in South-East Asia is **Indonesia**. Sumatra, Java, Kalimantan are the important oil producing regions of **Indonesia**. In China oil is produced in Sichuan, Shanxi, Yumen and Saidum which lie in the North-Western part. Crude oil is produced in 32 U.S. states and in U.S. coastal waters. In 2024 three maximum oil producing states was Texas, New Mexico and North Dakota. *In **Canada** petroleum is produced in Alberta province (Leduc region), Manitoba and British Columbia. *In the former Soviet Union petroleum was produced in the **Volga-Ural region, Baku region** (Azerbaijan) and regions lying in proximity the Caspian Sea. The country of **Venezuela** in South America is an important oil-producing country in the world. Petroleum exports constitute more than 80% (2024) **of the total exports of Venezuela**. Large scale production of petroleum is done in Brazil and Colombia. Normally mineral oil is found in those places where other fossil fuels are found. It is also found independently in some places. *According to the figures from the Indian Minerals Yearbook 2022, the top five leading countries in terms of natural gas reserves are – (1) **Russia** (2) **Iran**, (3) **Qatar** (4) **Turkmenistan** and (5) **USA**.

*The top five leading producers of natural gas in the world (2020) are – (1) **USA** (2) **Russia**, (3) **Iran**, (4) **China** (5) **Qatar**.

***Gasohol** is **90% Gasoline** (unleaded) and **10% ethyl alcohol** (ethanol) whereas gasoline or petrol is a mixture derived from petroleum. Hence, the largest producer and consumer of gasohol will be the one who produces a large amount of gasoline and ethanol. USA is the largest producer of Ethanol also in 2024. Hence, at present, USA is the **largest producer and consumer of gasohol**. ***Brent crude oil** is one of the classifications of crude oil and also the largest one. It comprises Brent Crude, Brent Sweet, Light crude, Oseberg Ekofisk and Fortis (BFOE). Brent Crude is extracted from the North Sea. It has a sulphur content of **0.37%**. Brent crude is used in the 2/3rd of the world's internationally trade crude oil supplies.

*Sugarcane, Maize, Soyabean, Rapeseed and Jatropha is used for producing Bio-fuels. The alcohol produced by sugarcane and Maize can be mixed with petroleum to produce Gasohol. Thus, reducing the problem of fossil fuel pollution. The oil extracted from Jatropha can be mixed with diesel to produce **Biodiesel**, which could be used as an alternative to diesel.

*The Gas pipeline between Iran and Pakistan has been given the name Peace Pipeline.

1. In relation to the production of petroleum the correct ascending order of the countries is –

- (a) China, Iran, U.S.A., Saudi Arabia
- (b) Iran, U.S.A., China, Saudi Arabia
- (c) Saudi Arabia, U.S.A. Iran, China
- (d) U.S.A. Iran, China, Saudi Arabia

U.P. Lower Sub. (Pre) 1998

Ans. (a)

Option (a) was correct when this question was asked. However presently, according to World Mineral Production 2018-22 (Data - 2022) leading petroleum producers are –

1. USA	-	759.50 Million tonnes
2. Saudi Arabia	-	573.10 Million tonnes
3. Russia	-	548.50 Million tonnes
4. Canada	-	274.00 Million tonnes
5. Iraq	-	221.30 Million tonnes

2. With reference to 'Alaska' which of the following statements is/are correct?

- (1) It is oil producing state of United States of America.
- (2) It was purchased from Russians in 1867.

Select the correct answer using the code given below:

Code -

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

'Alaska' is the oil producing state of USA. It was bought by America from Russia in 1867 AD.

3. The three leading producers of petroleum in the world in descending order are –

- (a) Russia, Saudi Arabia, U.S.A.
- (b) U.S.A., Saudi Arabia, Iran
- (c) Saudi Arabia, Russia, U.S.A.
- (d) Saudi Arabia, U.S.A., Russia

U.P.P.C.S. (Mains) 2008

Ans. (*)

According to World Mineral Production 2018-22, the three leading producers of petroleum in the world (2022) are USA, Saudi Arabia and Russia.

4. The largest petroleum producer in the world is –

- (a) Kuwait
- (b) Iran
- (c) Saudi Arabia
- (d) Russia

U.P.P.S.C. (GIC) 2010

Ans. (d)

In the year 2010, when this question was asked, the correct answer was option (d) but at present according to World Mineral Production 2018-22 (Data - 2022), Saudi Arabia is the second largest petroleum producer and the USA ranked first.

5. The correct sequence of leading producers of petroleum, in the Middle East in descending order is

- (a) Iran, Iraq, Kuwait, Saudi Arabia
- (b) Saudi Arabia, Iran, Iraq, Kuwait
- (c) Saudi Arabia, Kuwait, Iraq, Iran
- (d) Saudi Arabia, Iran, Kuwait, Iraq

U.P. Lower Sub. (Spl) (Pre) 2008

Ans. (*)

According to World Mineral Production 2018-22 (Data-2022), leading petroleum producers in the Middle East are – Saudi Arabia, Iraq, UAE and Iran.

6. Which of the following statement is true about petroleum production ?

- I. 60% of the world's petroleum deposits are found in the Middle East.**
- II. Petroleum reserve in Alaska is certified equivalent to Texas.**
- III. U.S.A. is a prominent producer and importer for petroleum.**

Code :

- (a) I and II are true
- (b) II and III are true
- (c) I and III are true
- (d) All are true

U.P.P.C.S. (Pre) 1998

Ans. (d)

The countries of Middle East i.e, Iran, Iraq, Kuwait and Saudi Arabia have more than 60% of the world's petroleum Reserves, when this question was asked. So, statement I is true. According to Indian Minerals Year Book-2022, the Middle East has the largest share (48.3%) of the world's petroleum deposits. The United States has petroleum reserves in 32 states. The three states with the largest petroleum reserves are – Texas, Alaska and California. The petroleum deposits of Texas and Alaska are almost equal. Thus, statement II is also correct. Presently, there are more reserves in Alaska than Texas. According to World Mineral Production 2018-22 (Data - 2022), the USA is the largest producer of petroleum followed by Saudi Arabia. U.S.A. is a prominent importing country but not top importing country in world. Thus, statement III is also correct.

7. The largest producer of petroleum of Southeast Asia is:

- (a) Brunei
- (b) Indonesia
- (c) Malaysia
- (d) Myanmar

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (b)

According to World Mineral Production 2018-22 (Data-2022), Indonesia is the biggest petroleum producer in Southeast Asia. Sumatra, Java and Kalimantan are important Petroleum producing regions in Indonesia.

8. Venezuela is a large –

- (a) Banana Producer
- (b) Zinc miner
- (c) Heroin (Drug) producer
- (d) Petroleum production centre

R.A.S./R.T.S. (Pre) 1993

Ans. (d)

South American country Venezuela is among the leading petroleum producing countries. The economy of Venezuela is dominated by the petroleum sector. Oil/petroleum exports generate more than 80% (2024) of the country's total export revenue.

9. World's largest reserves of mineral oil are in

- (a) Venezuela
- (b) Iran
- (c) Saudi Arabia
- (d) Iraq

U.P. Lower Sub. (Pre) 2013

U.P.P.C.S. (Pre) 2012

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (a)

According to data of 'Indian Minerals Year Book-2022' published by Indian Bureau of Mines, the largest proved crude oil deposits (48.0 thousand tonnes) are found in Venezuela. Saudi Arabia with a 40.9 thousand tonnes deposit is second place.

10. Which country has the largest reserves of oil?

- (a) United States (b) China
(c) Russia (d) Venezuela
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

See the explanation of above question.

11. The largest petroleum reserves are found in :

- (a) Iraq (b) Iran
(c) Saudi Arabia (d) U.S.A.

U.P.P.C.S. (Mains) 2006

Ans. (c)

In the year 2006, when this question was asked, option (c) was the correct answer but in present Venezuela in first place. The five countries with the largest petroleum reserves, according to Indian Minerals Year Book - 2022 are as follows:

(Country)	Reserve deposit (thousand tonnes)
Venezuela	- 48.0
Saudi Arabia	- 40.9
Canada	- 27.1
Iran	- 21.7
Iraq	- 19.6

12. Which one of the following countries has the largest known reserves of oil :

- (a) Kuwait (b) Iran
(c) Iraq (d) Nigeria

U.P.P.C.S. (Pre) 2002

Ans. (b)

According to Indian Minerals Year Book - 2022, five countries with the largest reserves (of oil) are – Venezuela, Saudi Arabia, Canada, Iran and Iraq. Since; Venezuela Saudi Arabia and Canada are not given in the options, the correct answer would be Iran.

13. In West Asia, the highest petroleum producing country is :

- (a) Iraq (b) Iran
(c) Saudi Arabia (d) Lebanon

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

According to World Mineral Production 2018-22 (Data - 2022), Saudi Arabia is the largest oil producing country in West Asia. It is the second largest oil-producing country in the world after the USA.

14. Which country is the largest producer and consumer of gasohol ?

- (a) Brazil (b) Cuba
(c) India (d) U.S.A.

U.P.P.C.S. (Mains) 2011

Ans. (d)

Gasohol is a mixture of unleaded gasoline (90%) and ethanol (10%). Gasoline is a byproduct mixture of petrol. Presently, the USA is the top producer (in 2023) and consumer of gasohol.

15. In the context of global oil prices, 'Brent crude oil' is frequently referred to in the news. What does this term imply ?

1. It is a major classification of crude oil.
2. It is sourced from the North Sea.
3. It does not contain sulphur.

Which of the statements given above is/are correct?

- (a) 2 only (b) 1 and 2
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2011

Ans. (b)

Brent Crude is a major classification of crude oil serves as a major benchmark price for purchases of oil worldwide. This grade is described as light because of its relatively low density due to its low sulphur content. Brent Crude is sourced from the North Sea and comprises Brent Blend, Forties Blend, Oseberg and Ekofisk crudes.

16. A commercial deal christened "Peace Pipeline" has been signed recently in the first half of 2009. Identify the pipeline from the following list :

- (a) Russia to Europe (b) Azerbaijan to Turkey
(c) Myanmar to China (d) None of the above

U.P.P.C.S. (Mains) 2007

Ans. (d)

Iran–Pakistan gas pipeline is also known as the 'Peace Pipeline' or IP gas pipeline.

17. Consider the following statements:

Statement-I: Sumed pipeline is a strategic route for Persian Gulf oil and natural gas shipments to Europe.

Statement-II: Sumed pipeline connects the Red Sea with the Mediterranean Sea.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

I.A.S. (Pre) 2024

Ans. (a)

The Sumed pipeline (also known as the Suez-Mediterranean pipeline) is an oil pipeline in Egypt, running from the Ain Sokhna terminal in the Gulf of Suez, the northernmost terminus of the Red Sea, to offshore Sidi Kerir port, Alexandria in the Mediterranean Sea. It provides an alternative to the Suez Canal for transporting oil from the Persian Gulf region to the Mediterranean. The Suez Canal and the Sumed pipeline are strategic routes for Persian Gulf crude oil, petroleum products, and liquefied natural gas (LNG) shipments to Europe and North America. The Suez Canal and Sumed pipeline are both located in Egypt and connect the Red Sea with the Mediterranean Sea. Hence, both Statement-I and Statement-II are correct, and Statement-II explains Statement-I. So, option (a) is the correct answer. However, it is to be noted that Sumed pipeline is an oil pipeline which transports crude oil only (not natural gas). So, in literal terms, Statement-I could be incorrect and in that case option (d) will be the correct answer.

18. In which of the following countries mineral oil was commercially drilled in 1857 ?

- (a) U.S.A. (b) Venezuela
(c) Indonesia (d) Romania

R.A.S./R.T.S.(Pre) 2013

Ans. (d)

Romania was the country in the world with an oil production of 275 tonnes officially registered in 1857 in international Statistics. The first commercial oil well in Romania was drilled in 1857.

19. Which crop group out of the following contains crop/crops that cannot be used for biofuel production?

- (a) Sugarcane, Corn, Mustards
(b) Jatropha, Sugarcane, Palm
(c) Lentil, Sugarbeet, Wheat
(d) Soyabean, Corn Rapeseed

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (c)

Lentil, Sugarbeet and Wheat are crops that cannot be used for biofuel production. Biofuel is a fuel that is derived from biomass, that is plant material or animal waste. Sugarcane, Corn Soyabean, Rapeseed and Jatropha are used for biofuel production. Biofuel is cost effective and an alternative to petroleum.

20. Among the following countries, which was the highest crude oil supplier to India during 2017-18 :

- (a) Saudi Arabia (b) Iran
(c) Iraq (d) Kuwait

U.P.R.O./A.R.O. (Pre) 2017

U.P.P.C.S. (Mains) 2017

Ans. (c)

Iraq was the largest supplier of crude oil to India in the year 2017-18. It is worth mentioning that earlier Saudi Arabia was the largest supplier of crude oil, when this question was asked. According to Indian Minerals Year Book 2022 (Data - 2021-22 'P'), Iraq supplied maximum amount of petroleum (Crude) to India. It is followed by Saudi Arabia and United Arab Emirates (UAE) respectively.

21. Where is Ras Tanura Oil Refinery located?

- (a) Iran (b) United States of America
(c) Saudi Arabia (d) Iraq

M.P.P.C.S. (Pre) 2017

Ans. (c)

Ras Tanura is the one of the oldest refineries on the Persian Gulf coast, located near the industrial port city Jubail in Saudi Arabia. This oil refinery was started in 1945.

22. Which one of the following pairs (Country — Oil field) is not correctly matched?

- (a) Iran — Hafi Kel
(b) Kuwait — Kashagan
(c) Saudi Arabia — Dhahran
(d) Iraq — Zubair

UPPSC (Pre) 2024

Ans. (b)

Kashagan is a significant and vast oil and gas field in Kazakhstan. The Kashagan oil Field Refinery is located here, which is one of the world's leading companies in gas production, petroleum products, petrochemicals and their refining.

23. Which one among the following countries of the world, except the United States of America, is the largest crude oil producer?

- (a) Russia (b) China
(c) Saudi Arabia (d) Canada
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

According to World Mineral Production 2018-22 in the year 2022 USA was the largest crude oil producer followed by Saudi Arabia.

24. Which among the following is an important mineral of Pegu Yoma region of Myanmar?

- (a) Tin (b) Silver
(c) Mineral oil (d) Copper

U.P.P.C.S. (Pre) 2022

Ans. (c)